MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS) BASE MATERIAL GX, GLASS BASE, WOVEN, POLYTETRAFLUOROETHYLENE RESIN, FLAME RESISTANT, FOR MICROWAVE APPLICATION, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values. See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.030 inch to 0.010 inch	0.030 inch and over
1. Peel strength, minimum in pounds per-inch width			
A. 1/2 oz/ft^2 copper			
As received	A	4.0	4.0
After thermal stress	A	4.0	4.0
At elevated temperature	150°C	2.0	4.0
After exposure to processing solutions	A	4.0	4.0
B. 3/4-oz/ft^2 copper			
As received	A	4.0	4.0
After thermal stress	A	4.0	4.0
At elevated temperature	150°C	2.0	4.0
After exposure to processing solutions	A	4.0	4.0
C. 1-oz/ft^2 copper			
As received	A	5.0	8.0
After thermal stress	A	5.0	8.0
At elevated temperature	150°C	2.0	8.0
After exposure to processing solutions	A	5.0	8.0
D. 2-oz/ft^2 copper			
As received	A	6.0	10.0
After thermal stress	A	6.0	10.0
At elevated temperature	150°C	2.0	10.0
After exposure to processing solutions	A	6.0	10.0
E. 3-, 4-, and 5-oz/ft^2 copper			
As received	A	7.0	12.0
After thermal stress	A	7.0	12.0
At elevated temperature	150°C	2.0	12.0
After exposure to processing solutions	A	7.0	12.0

See footnotes at end of table.

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TABLE I Property values - Continued.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.030 inch to 0.010 inch	0.030 inch and over
2. Volume resistivity, minimum in megohm-centimeter			
After moisture resistance	F	10^6	10^6
At elevated temperature	E-24/150	10^5	10^5
3. Surface resistivity, minimum megohm			
After moisture resistance	F	10^4	10^4
At elevated temperature	E-24/150	10^3	10^3
4. Water absorption, maximum percent	E-1/105+des +D-24/23	0.010-0.20 0.020-0.20	0.030-0.20 0.060-0.10 0.090-0.08 0.120-0.08 0.240-0.05
5. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0 5/23	N/A	20
6 Permittivity at X band 2/ 3/	C-24/23/50	2 40 ±.04 4/ 2.45 ±.04 5/ 2.50 ±.04 2.55 ±.04 2.60 ±.04	2.40 ±.04 2.45 ±.04 2.50 ±.04 2.55 ±.04 2.60 ±.04
7. Loss tangent at X band 2/	C-24/23/50	0.0023 5/	0.0022
8 Q (resonance), minimum	N/A	N/A	N/A
9. Flexural strength, minimum thousand lb/in^2			
Lengthwise	A	N/A	12.0
Crosswise	A	N/A	10.0
Lengthwise at elevated temperature	N/A	N/A	N/A
10 Arc resistance, minimum seconds	D-48/50+ D-0.5/23	120	180
11. Flammability, maximum burntime, seconds, each flame application 6/	N/A	N/A	N/A

1/ See specimen conditioning requirements, conditioning tolerances, and testing information table XXVI of MIL-P-13949.
2/ For the applicable test method (see IPC-TM-650, method 2.5.5.5).
3/ Value shall be specified in ordering data.
4/ Tolerance of ±0.02 (0.51 mm) for all values of dielectric constant may be specified in the ordering data.
5/ Materials other than 0 030 (0 76 mm) and 0 060 (1.52 mm) inch thick are not testable at 10 GHz.
6/ PTFE materials will not burn in small scale flame tests

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Thickness. Cut sheets less than 24 by 14 inches shall meet the applicable thickness tolerances in 100 percent of the area of the sheet.

Marking. Internal marking not applicable.

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent
DLA - ES

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